MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
RELIABILITY & CONDITION MONITORING CELL

ROTATING MACHINERY VIBRATION & SEAL DIAGNOSTIC REPORT
REPORT NO: **MRPL-RMC-VIB-2026-114** | DATE: 02-MAR-2026 | ASSET: **P-101B (STANDBY)**

1. VIBRATION ANALYSIS DATA SUMMARY (ISO 10816-3)

Measurement Point	Direction	Overall RMS (mm/s)	Velocity Peak (mm/s)	Dominant Harmonic	Zone Evaluation
Pump DE Bearing	Horizontal	4.82	6.12	1X RPM (24.7 Hz)	Zone B (Acceptable)
Pump DE Bearing	Vertical	3.20	4.05	1X RPM	Zone A (Good)
Pump DE Bearing	Axial	5.85	7.40	1X + 2X RPM	Zone C (Unsatisfactory)
Pump NDE Bearing	Horizontal	7.65	9.80	2X Vane Pass (148 Hz)	Zone C (RESTRICTED RUN)
Motor DE Bearing	Horizontal	2.10	2.80	1X RPM	Zone A (Good)

2. MECHANICAL SEAL & ANCILLARY INSPECTION

• High axial vibration on NDE bearing correlates with acoustic emissions indicating hydraulic recirculation.
• Mechanical seal cartridge (API Plan 53B) observed with barrier fluid pressure drop (3.5 bar → 2.1 bar over 48h).
• Pinhole corrosion pitting noted on seal flush recirculation tubing adjacent to orifice FO-101B.
• **Action**: P-101B shall not be placed in continuous service. Overhaul seal assembly and replace flush tubing.

Vibration Analyst: K. M. Prabhu

[Recorded & Analysed 02/03/2026]

Maintenance Superintendent: D. A. Gowda

[Action Plan Approved 03/03/2026]